

EcoTrace India

Product Requirement Document (PRD) • Version 3.0

Product Name	EcoTrace India (Web-Based Carbon Tracking Ecosystem)	Status	Ready for Review
Author	Sridhar Chandrasekaran	Date	June 11, 2026
Platform Target	Web App (Responsive Progressive Web App via Firebase)	Target Audience	Indian Urban Households, Professionals, & Academic Communities

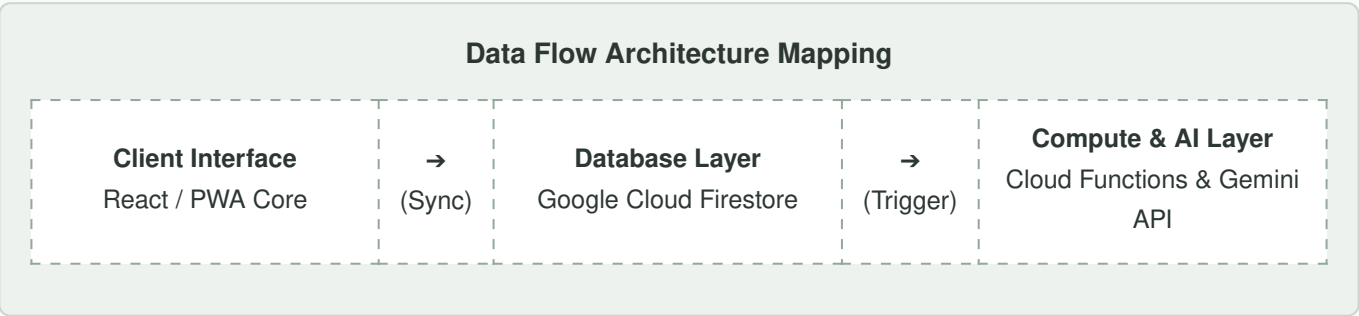
1. Executive Summary & Core Mission

EcoTrace India is a localized, high-performance web application designed to track, analyze, and mitigate personal and household carbon emissions across the Indian demographic. Version 3.0 deepens our technical architecture by nesting the application entirely within the **Google Cloud and Firebase Ecosystem** to ensure hyper-scalability, near-zero server management, and minimal latency across tier-1, tier-2, and tier-3 networks in India.

Additionally, this version introduces the **EcoAgent AI Layer**—an embedded, contextual intelligence panel natively integrated into the user workflow. Rather than serving static analytics and passive charts, the interface displays an interactive AI Agent column that interfaces directly with real-time user metrics, answers targeted queries from candidates/participants, interprets complex emission logs, and delivers immediate, personalized mitigation choices dynamically.

2. Technical Architecture & Google Ecosystem Blueprint

To ensure architectural cleanliness, optimal resource consumption, and frictionless validation, the web application relies explicitly on Google's fully managed infrastructure suite:



2.1 Storage & Synchronization: Google Cloud Firestore

- **NoSQL Document Structure:** User configurations, session profiles, and emission ledgers are stored in hierarchical, schema-validated collections (e.g., `users/{userId}/footprints/{monthId}`).
- **Real-Time SDK Realization:** Client-side interfaces instantiate Firestore's real-time listeners via `onSnapshot`. Updates to utility logs, transport metrics, or meal plans instantly trigger updates to the domestic carbon weight on the client view without triggering standard REST overhead or excessive compute resource consumption.
- **Offline Persistence Matrix:** Network fault tolerance is active by default. Writes initiated on erratic regional transits or underground subway routes cache directly in indexedDB, completing server synchronization once standard connection parameters return.

2.2 Security, Deployment, & Operations

- **Authentication Gateway:** Managed securely via **Firebase Authentication**, utilizing localized Indian workflows including Google Sign-In and low-latency Phone/OTP validation modules matching domestic consumer trends.
- **Granular Firestore Security Rules:** Role and path isolation are strictly enforced at the database root to prevent data bleed across user identities:

```
match /users/{userId}/{document=**} {
  allow read, write: if request.auth != null && request.auth.uid == userId;
}
```

- **High-Speed Hosting Infrastructure:** Deployed via **Firebase Hosting** using global CDNs pinned with local regional Edge points in Mumbai and Delhi to optimize Time to First Byte (TTFB) over standard mobile platforms.

3. High-Level Feature Scope & Component Layout

Epic 1: Multi-Tab Localized Carbon Tracking Matrix

The layout partitions active operational tools into distinct thematic tracking matrices (Diet, Transport, Utilities, Infrastructure). Each tab leverages a unified layout paradigm:

The Entry & Ledger View (Left 65%)	The EcoAgent Panel (Right 35%)
Houses specific localized field logs. Inputs match day-to-day Indian paradigms: kWh/INR from state electricity boards (e.g., BESCO, MSEB), LPG cylinder depletion cadences, and multi-modal travel variables (Metro, local trains, auto-rickshaws, shared two-wheelers).	A persistent conversational companion column locked alongside input states. It processes contextual information from the active matrix layout to address candidate questions without requiring page redirection.

Epic 2: The EcoAgent Intelligence Column

- **FR-2.1 (Contextual Awareness):**** The Agent continuously reads the state mutation arrays directly from the user's active Firestore documents for that open session.
- **FR-2.2 (Candidate Inquiries):**** Candidates can execute raw conversational text lookups or engage quick-action tokens (e.g., *"Why is my transport footprint exceeding the Bengaluru municipal benchmark?"* or *"Calculate my cost savings if I switch my AC baseline to 24°C."*).
- **FR-2.3 (Automated Response Dispatch):**** Conversational processing runs asynchronously via HTTPS triggers running over Node-based Firebase Cloud Functions paired directly with the **Gemini API (Vertex AI)**, delivering validated guidance tuned to Indian economic and geo-specific data.

4. Evaluation & Design Standards Matrix

The structural and operational codebase undergoes evaluation against five mandatory deployment vectors:

4.1 Cleanliness, Readability, & Structural Integrity

UI rendering elements must remain detached from pure carbon coefficient formulas. Constants mapping Indian grid constraints (e.g., 0.82 kg CO₂e per kWh) must reside within isolated utility files. Structural definitions must use typed TypeScript models to reject mismatched database schemas.

4.2 Security Vulnerability Prevention & Safe Practices

All inputs bound for the EcoAgent pipeline must clear regex filtering to completely mitigate prompt-injection vectors and Cross-Site Scripting (XSS). Administrative balance mutations, metric weights, and public leaderboard logs must settle strictly via server-side Cloud Functions.

4.3 Efficiency & Resource Optimization (Time & Memory)

Implement selective memoization arrays (e.g., React's useMemo) to prevent recursive renders and redundant Firestore document snapshot assessments. The absolute network bundle size must leverage route-based split lazy loading, maintaining initial core bundle distribution under 150KB for consistent 4G/5G mobile executions.

4.4 Testability, Validation, & Maintainability

Core computation modules require 100% test verification patterns using isolated frameworks (Jest/Vitest). Local deployments must enforce simulation verification via the **Firebase Local Emulator Suite** before staging code updates to the main repository.

4.5 Usability Across Diverse Environments

The interface must dynamically stack, converting the split 35% EcoAgent panel into a clean bottom-drawer sheet when viewing on mobile screen layouts. Asset assets must load as WebP formats alongside strict cache control indicators to preserve restricted mobile bandwidth caps.